

36. Case Study 1: The Auditorium

In a school auditorium, seats are arranged in rows such that the number of seats in each row increases by a fixed number. There are 30 seats in the first row, 34 in the second, 38 in the third, and so on.

- (i) Find the number of seats in the 10th row. (1 Mark)
- (ii) If there are 15 rows in the auditorium, what is the total seating capacity? (1 Mark)
- (iii) **(a)** Find the total production in the first 7 years. (2 Marks) **OR**
(b) How many seats are there in the middle row if there are 25 rows in total? (2 Marks)